100 DAYS OF MACHINE LEARNING

DAY 02

Batch Learning • Online Learning
Instance-Based • Model-Based Learning

Complete Notes & Practice Workbook

Beginner-friendly explanations, comparison tables, real-world examples, Python practice, exercises, worked answers and revision checklist.

Learning rule: Understand the idea, then practise it. A quiz score is a useful checkpoint; practical examples help build deeper understanding.

Day 2 learning outcomes

By the end of this lesson, you should be able to explain batch and online learning, distinguish instance-based from model-based learning, recognise KNN and Linear Regression as examples, and choose a reasonable approach for a simple scenario.

1. Batch Learning

Definition: Batch learning trains a model on a collected dataset in a separate training process. When new information needs to be included, the model is often retrained on an updated dataset.

Example — university marks: A university collects student records from several semesters and trains a model to estimate marks from features such as attendance, assignment scores and study hours. At the end of a new semester, the team may add the latest records and train a new version.

Advantages: Training can be planned and tested in a controlled environment; data and model versions can be tracked.
Limitations: New patterns may not be reflected until the next training cycle; retraining can be costly for very large datasets.

Roman Urdu: Batch learning mein pehle data collect hota hai, phir model ko us dataset par train kiya jata hai. Naya data include karne ke liye aksar model next training cycle mein dobara train hota hai.

2. Online Learning

Definition: Online learning updates a model incrementally as new data arrives. Depending on the algorithm, an update may happen for each record or for a small mini-batch.

Example — changing sales: An online shop receives fresh sales data throughout the day. A suitable online-learning system may update its model as data arrives, helping it adapt to changing customer behaviour.

Advantages: New information can be incorporated quickly; useful for streams or very large datasets; updates need not always retrain from scratch.
Limitations: Noisy data can harm the model; monitoring and careful update settings matter; concept drift can reduce performance.

Important: Online does not guarantee that every record triggers an immediate update. Some systems process mini-batches. It also does not mean the model must be connected to the internet.

Batch vs Online — comparison

Feature	Batch learning	Online learning
Training style	Train on collected data in a separate cycle.	Update incrementally from incoming data or mini-batches.
New data	Often waits for the next retraining cycle.	Can be incorporated sooner.
Compute pattern	May require a larger planned training job.	Updates can be smaller and more frequent.
Main concern	Model may become stale between cycles.	Noisy data or poor updates can hurt performance.
Example	Monthly sales forecast retraining.	Incrementally updated sales model.

3. Instance-Based Learning

Definition: Instance-based learning predicts a new case by comparing it with stored training examples. The method depends on how similarity or distance is defined. K-Nearest Neighbors (KNN) is a well-known example.

Student example: Training examples are (study hours, marks): (2, 40), (4, 60), and (6, 80). For a new student who studies 4.5 hours, a simple nearest-neighbor method finds the closest study-hours value. Four hours is closest, so a one-neighbor method returns 60 marks.

Study hours	Observed marks
2	40
4	60
6	80

This is a tiny teaching example, not a reliable marks predictor. Real systems need representative data, relevant features and evaluation on unseen data.

Advantages: Simple to understand; can represent local patterns; new examples can sometimes be added without fitting a new global equation.
Limitations: Prediction can be slow with many examples; memory use can be high; distance metrics, feature scaling and neighbour choice matter.

Roman Urdu: Instance-based approach naye case ko puranay examples ke saath compare karti hai. Similar examples ki information prediction mein madad karti hai.

4. Model-Based Learning

Definition: Model-based learning uses training data to learn a general pattern, relationship or set of parameters. Predictions are then made using the learned model.

Example — house prices: A Linear Regression model may learn an approximate relationship between house size and price. It can then estimate a price for a new house size.

A simple linear model can be written as:

predicted price = slope × house size + intercept

The slope and intercept are learned from data. Real house prices also depend on location, age, condition and market changes.

Advantages: Predictions can be efficient after fitting; a general pattern can be compactly represented.
Limitations: The model may oversimplify reality; assumptions may not fit; performance depends on data quality and evaluation.

Roman Urdu: Model-based learning data se ek general rule ya pattern seekhti hai. Phir us rule ko use karke naye examples ki prediction karti hai.

Instance-Based vs Model-Based — comparison

Feature	Instance-based	Model-based
Core idea	Compare with stored examples.	Learn a general pattern or function.
Common example	K-Nearest Neighbors (KNN).	Linear Regression.
Prediction	May search for similar examples.	Uses learned parameters or model structure.
Potential drawback	Can be slow and memory-intensive.	A poorly chosen model may miss patterns.
Useful when	Local similarity is informative.	A general relationship can be learned.

5. These are different dimensions

A common beginner mistake is to treat batch/online and instance/model-based as competing labels on one scale. They describe different aspects of learning.

Question	Concept pair	What it describes
When/how is the model updated?	Batch vs online	Training and update schedule.
How does the method make predictions?	Instance-based vs model-based	Prediction strategy and what is learned or retained.

A system can combine ideas from more than one category. An instance-based method can receive new examples over time; a model-based method can be retrained in batches or updated incrementally if its algorithm supports it.

6. How to choose an approach

Scenario	Possible choice	Reasoning
University updates a predictor each semester.	Batch learning	A planned retraining cycle may be sufficient.
New records must influence a model quickly.	Online learning	Incremental updates may fit, with monitoring.
Similar customers inform a recommendation.	Instance-based	Similarity can identify comparable cases.
Features have a learnable relationship with price.	Model-based	A fitted model can capture a general relationship.

These are examples, not absolute rules. The right choice depends on dataset size, update frequency, latency, memory, interpretability, cost and evaluation.

7. Python practice

The accompanying Python file demonstrates a tiny nearest-neighbour prediction and, if scikit-learn is installed, fits a Linear Regression model. It prints the key distinctions between the learning approaches.

Install optional packages:

```
pip install scikit-learn numpy
```

Run the practice file:

```
python CampusX_ML_Day_02_Complete_Practice.py
```

What to notice

• Nearest-neighbour prediction returns the target of the closest stored example.
• Linear Regression learns coefficients and predicts with a fitted relationship.
• A plausible prediction does not prove generalization. Evaluate on suitable data not used during training.

8. Exercises

1. A bank retrains its risk model every Sunday using collected records. Which training style does this resemble?
2. A sensor system updates its model using small chunks of new data. Which training style could fit?
3. A method predicts preferences by comparing similar customers in stored data. Instance-based or model-based?
4. A model learns coefficients for a formula estimating delivery time. Instance-based or model-based?
5. Explain one risk of online learning and one risk of batch learning.

9. Worked answers

1. Batch learning, because the model is retrained on a collected set of records on a planned schedule.
2. Online learning, because it updates incrementally from incoming mini-batches.
3. Instance-based learning, because it compares the new customer with similar stored examples.
4. Model-based learning, because the model learns parameters for a general formula.
5. Online learning can be harmed by noisy incoming data or concept drift. Batch learning may not reflect new patterns until the next retraining cycle.

10. Common misconceptions

• **“Online means internet-connected.”** Not necessarily; it describes incremental model updates.
• **“Online always updates after every record.”** Not necessarily; mini-batches can be used.
• **“Instance-based means no learning occurs.”** It still uses examples and may need data preparation; the prediction strategy differs from fitting a global model.
• **“Model-based is always better.”** No approach is best for every problem.
• **“A plausible prediction means the model is accurate.”** Accuracy needs suitable evaluation data and metrics.

11. Revision checklist

■ Batch learning trains on collected data in a planned cycle.
■ Online learning updates incrementally from new data or mini-batches.
■ Online systems need monitoring for noise and changing patterns.
■ Instance-based learning uses similarity to stored examples.
■ Model-based learning learns a general relationship or structure.
■ KNN is a common instance-based example.
■ Linear Regression is a common model-based example.
■ Batch/online and instance/model-based describe different dimensions.

12. Day 2 completion record

Quiz result: 5/5 (100%), as reported after the Day 2 understanding check.

Next step: Review these examples and run the Python practice file. We will begin the next official playlist lesson after confirming the core ideas remain clear.

Study source: CampusX — 100 Days of Machine Learning official playlist:
https://youtube.com/playlist?list=PLKnIA16_Rmvbr7zKYQuBfsVkjoLcJgxHH

Prepared as a beginner study workbook. Examples are simplified and are not production-ready ML systems.